

A ONE-LINE SEPARATION CRITERION FOR ISOTROPY QUOTIENTS OF CUBICAL SITES WITH REVERSAL

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ABSTRACT. On the De Morgan, Kleene, and Boolean cubical sites, the cubical-type and test model structures classify isotropy quotients $\square^n/H$ into agreement and separation cases; in dimensions ≤ 3 this was settled by a case-by-case classification with four agreement families and three separation mechanisms. This paper replaces the case analysis by a single criterion, uniform in the dimension and in the site. For a mixed subgroup H of the hyperoctahedral group, let N be the normal closure of the reflections of H (the elements with fixed cells). Then: the quotient agrees if N still has a common fixed cell, and carries an attached separating map if N is itself mixed. The agreement half is proved unconditionally by a two-step covering argument: N acts trivially on its fixed locus, so an unsigned consensus retraction contracts $\square^n/N$ in both structures, and H/N — which contains no reflections — acts freely, so the quotient is a classifying object $B(H/N)$ in both structures. The separation half rests on an exact characterization of the certificate method: the obstruction certificates of the classification papers succeed for a group precisely when it is generated by its reflections, because consistency of the certificate systems forces every twist to kill all vertex stabilizers, every reflection fixes a vertex, and conversely any nontrivial quotient of the reflection closure supports an explicit multiplexer cell. Since N is by construction generated by reflections, it is certified; the attached separation then descends to H along the free quotient by the covering theory of the companion paper. In dimension ≤ 3 the criterion reproduces the known 78/20 classification with zero mismatches, and all statements are machine-verified; in dimension 4 the new primitive groups are certified as predicted.

1. INTRODUCTION

Let $\square_\gamma$ be one of the three standard cube categories with reversal — De Morgan, Kleene, Boolean — and consider, on its presheaf site, the cubical-type model structure of cubical type theory and Cisinski’s test structure [1, 2, 4, 7]. The two structures are separated [8, 6], and the natural finite testing ground for *how* they differ is the family of isotropy quotients $\square^n/H$, $H \leq B_n$ the hyperoctahedral group. In dimensions $n \leq 3$ on the De Morgan

2020 *Mathematics Subject Classification.* 55U35; 18N40, 20F55, 06D30, 08B20.

Key words and phrases. cubical sets, cubical type theory, model structure, isotropy quotient, reflection group, covering space, hyperoctahedral group.

site, the classification paper [3] sorted all 98 subgroups into 78 agreement cases (four families: coverings, fixed cells, median retractions, products) and 20 separation cases (three mechanisms: parity certificates, block transfer, homological obstructions), and the landscape paper [5] transported the classification to the Kleene and Boolean sites (same conclusions, provably different proofs).

This paper replaces all seven families and mechanisms by one criterion.

Definition 1.1. An element of B_n is a *reflection* if it is nontrivial and fixes a nonempty cell of $\square^n$; equivalently, every cycle of the underlying signed permutation has even sign sum. A subgroup H is *mixed* if it contains a reflection but no nonempty cell is fixed by all of H . Write $\text{Refl}(H)$ for the set of reflections of H and $\langle\langle \text{Refl}(H) \rangle\rangle$ for the normal closure of the subgroup they generate.

Theorem 1.2 (The criterion). *Let $H \leq B_n$ be mixed and set $N := \langle\langle \text{Refl}(H) \rangle\rangle$. On each of the De Morgan, Kleene, and Boolean sites:*

- (1) *If N has a common fixed cell, then $\square^n/H$ has the same homotopy type in the type and test structures: both are classifying objects $B(H/N)$.*
- (2) *If N is mixed, then an explicit test weak equivalence that is not a type weak equivalence is attached to H (under Hypothesis 5.2, automatically verified for $n \leq 3$).*

In dimension ≤ 3 the criterion reproduces the 78/20 classification of [3] exactly.

The two halves are proved by two arguments that are, individually, simpler than any of the mechanisms they replace.

Agreement: the two-step covering. If N still fixes a cell, its fixed locus is a subcube $L \cong \square^d$, and — the key point — N acts *trivially* on L . Consequently the consensus retraction of $\square^n$ onto L (collapse each N -class of coordinates to an unsigned unanimity term such as their meet) is N -equivariant with no sign conditions at all, and $\square^n/N \simeq L$ is contractible in both structures. The quotient group H/N contains no reflections — all of H 's reflections were swallowed by N — and a group element fixing a cell class would produce a new reflection, so H/N acts *freely* on the cells of $\square^n/N$. By the covering theory of [5], a free quotient of an object contractible in both structures is a $B(H/N)$ in both structures. This subsumes, with one proof, the fixed-cell, median, and agreeing-product families of [3]: the median retraction, whose odd-arity self-dual majority term had seemed essential, is revealed to be an avoidable device — performing the retraction at the level of N , where no signs act, and only then passing to the free cover, eliminates the equivariance problem that the median was solving.

Separation: certificates are exactly reflection-generation. The separation half rests on an exact characterization of when the obstruction-certificate method of [3, 5] succeeds.

Theorem 1.3 (Certificates = reflection generation). *For a mixed H with witness machine as in [3], the exact certificate system (no consistent non-trivial twist) holds if and only if H is generated by the normal closure of its reflections.*

The proof identifies the mechanism of the certificates: a consistent twist yields an equivariant cell whose vertex map, combined with the free vertex system, forces the twist to kill every vertex stabilizer; nontrivial elements fixing a vertex are reflections, and every reflection fixes a vertex, so the twist kills $\langle\langle \text{Refl}(H) \rangle\rangle$. Conversely, any nontrivial cyclic quotient of $H/\langle\langle \text{Refl}(H) \rangle\rangle$ supports an explicit consistent twist by a multiplexer cell. Since $N = \langle\langle \text{Refl}(H) \rangle\rangle$ is generated by reflections *by construction*, it is always certified; the machinery of [3] (in the multi-orbit form recalled in Section 5) attaches a separating map to N , and the covering-descent theorem of [5] transports it to H along the free quotient H/N . The same vertex-fixing argument shows that the auxiliary hypothesis of [3] — triviality of realizable characters — is automatic for reflection-generated groups, on every site: a realizable character must vanish on vertex stabilizers, hence on all reflections, hence on N .

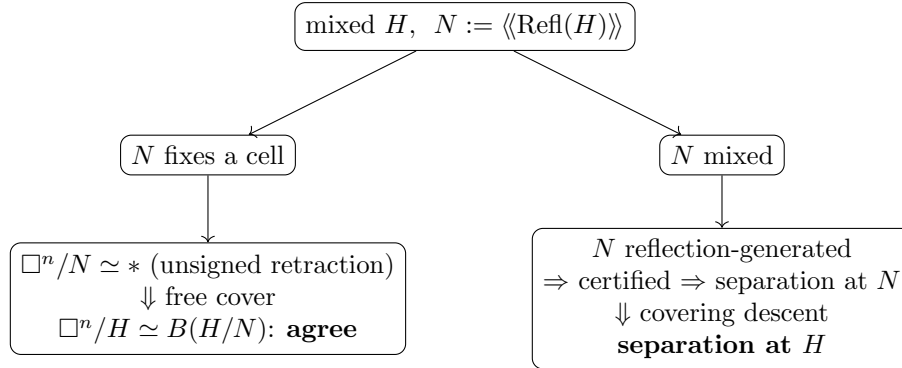


FIGURE 1. The criterion and its two proofs.

Relation to the companion papers. This is the fourth paper of the series on the site-dependence of the type/test comparison: [8] produced the separation, [6] the phase diagram over the interval theories, [3] the three-dimensional classification, [5] the transport of the classification across the reversal sites together with the covering machinery (the equivariant replacement of free quotients and the descent of non-equivalence) that the present paper uses as its main engine. The conceptual yield here is that the classification, originally a case analysis, is a theorem about reflection subgroups:

everything that happens to an isotropy quotient is controlled by how much of its symmetry is generated by reflections.

2. PRELIMINARIES

We freely use the setting of [3, 5]: the three reversal sites and their free-algebra representations; the cubical-type and test structures; witness machines (the join collage for two-stratum groups, the E -resolution collage in general) with their twist calculus; and the covering theory of [5]: presheaf coverings are local systems on the category of elements, coverings extend uniquely along the algebraic fibrant replacement (the box domains are connected), free quotients admit equivariant replacements, and a map of free quotients is a type weak equivalence if and only if its equivariant cover is.

Two elementary facts about reflections anchor everything.

Lemma 2.1. *(i) Every reflection fixes a vertex of $\square^n$. (ii) Every nontrivial element fixing a vertex is a reflection.*

Proof. (i) On a cycle of even sign sum, propagate a vertex value around the cycle: the total twist is the sign sum, which is even, so a compatible assignment exists on each cycle. (ii) A vertex is a nonempty cell. $\square$

Lemma 2.2 (Automatic character triviality). *Let ψ be a character of H realizable as a twisted element of the free algebra on any of the three sites ($T \circ \sigma_h = \neg^{\psi(h)} T$). Then ψ vanishes on every reflection, hence on $\langle\langle \text{Refl}(H) \rangle\rangle$. In particular reflection-generated groups have no nontrivial realizable characters, on any reversal site.*

Proof. Evaluating the twisted equation at a vertex v fixed by h gives $T(v) = \psi(h) \oplus T(v)$ pointwise (Boolean; on the De Morgan and Kleene sites evaluate at the corresponding proper literal point, which is ρ -compatible), so $\psi(h) = 0$ for h in a vertex stabilizer. Apply Lemma 2.1(i). $\square$

3. CERTIFICATES ARE EXACTLY REFLECTION GENERATION

We work with the exact certificate system of [5]: for a mixed H with a reflection-free normal $V \trianglelefteq H$ and $G = H/V$, and a twist $\delta \in \text{Hom}(H, H)$, the u -part asks for an equivariant cell $c : \square^n \rightarrow \square^n$ with $c \circ \sigma_h = \delta(h) \circ c$, and the e -part for a G -valued vertex function E with $E(h \cdot v) = \bar{\delta}(h) \cdot E(v)$; on the Boolean site both systems are exact (parity-consistent assignments are cells), and on the Kleene site exactness holds after the monotone-realizability filter. H is *certified* when no nontrivial δ is consistent.

Theorem 3.1. *H is certified if and only if $H = \langle\langle \text{Refl}(H) \rangle\rangle$.*

Proof. ($\Leftarrow$) Let δ be consistent. The u -part cell restricts to a vertex map $\hat{c}$ with $\hat{c}(h \cdot v) = \delta(h) \cdot \hat{c}(v)$, so $\delta(\text{Stab}_H(v)) \subseteq \text{Stab}_{B_n}(\hat{c}(v))$. The e -part, since G acts freely on itself, forces $\bar{\delta}(\text{Stab}_H(v)) = 1$, i.e. $\delta(\text{Stab}_H(v)) \subseteq V$. But V is reflection-free while nontrivial vertex stabilizers consist of reflections

(Lemma 2.1(ii)), so $\delta(\text{Stab}_H(v)) = 1$ for every vertex v . By Lemma 2.1(i) every reflection lies in some vertex stabilizer, so δ kills $\text{Refl}(H)$, hence the normal closure ($\ker \delta$ is normal), hence H ; thus δ is trivial.

($\Rightarrow$) Suppose $N := \langle\langle \text{Refl}(H) \rangle\rangle \subsetneq H$. Choose a nontrivial character $\psi : H \rightarrow C$ onto a cyclic group vanishing on N (a nontrivial quotient of the finite group H/N onto a cyclic group exists), and $k \in H$ with $\psi(k)$ generating C and $|k|$ -compatible; the vertex action of B_n is faithful, so k moves some vertex v_0 . Since ψ vanishes on all vertex stabilizers (they consist of reflections), there is a ψ -twisted vertex-level function $T : \{0, 1\}^n \rightarrow C$ with $T(h \cdot p) = \psi(h) + T(p)$ (propagate along orbits; consistency at stabilized points is the vanishing). The *multiplexer cell* $c(p) := k^{T(p)} \cdot v_0$ — encoded coordinatewise as a Boolean cell — satisfies $c(\sigma_h p) = \delta(h) \cdot c(p)$ for $\delta(h) := k^{\psi(h)}$, which is a homomorphism because k has order divisible by $|C|$ along ψ ; and $E(h \cdot v_0) := \bar{\delta}(h)$ is well-defined on the orbit of v_0 and extends freely elsewhere. Hence δ is a consistent nontrivial twist and H is not certified. $\square$

Remark 3.2. Theorem 3.1 explains the empirical tables of [5] at a glance (14 groups across $n = 2, 3, 4$, all matching), identifies the De Morgan certificates of [3] as instances (there the realizability filter is strictly stronger, but reflection-generation still implies certification by the same argument, and Lemma 2.2 discharges the character hypothesis), and turns “certified” from a computation into a group-theoretic property.

4. THE AGREEMENT HALF

Theorem 4.1 (Two-step covering agreement). *Let H be mixed and suppose $N = \langle\langle \text{Refl}(H) \rangle\rangle$ has a common fixed cell. Then, on each reversal site, $\square^n/N$ is contractible in both structures, H/N acts freely on its cells, and*

$$\square^n/H \simeq B(H/N)$$

in both structures. In particular $\square^n/H$ agrees.

Proof. Step 1. The common-fixed-cell condition for N is a consistent parity system on the coordinates, whose classes and signs define the fixed locus $L \cong \square^d$ with $d < n$ (a nontrivial reflection has a cycle of length ≥ 2 , so some class has size ≥ 2). N acts *trivially* on L : an element of N fixes every cell of L by definition of the common fixed pattern. Consequently the induced action of N on the classes carries no signs, and the consensus retraction

$$(R_t)_i = F(x_i, \neg^{\varepsilon_i} M_{\text{cl}(i)}, t), \quad M_k := \bigwedge_{j \in \text{class } k} \neg^{\varepsilon_j} x_j,$$

with F the median interpolation of [8], is N -equivariant — the only equivariance conditions are permutations *within* and *among* equal-size classes without signs, which the symmetric M satisfies — and no self-duality is

required anywhere. R descends to the quotient and contracts $\square^n/N$ onto $L/N = L \cong \square^d$, strictly; hence $\square^n/N$ is contractible in both structures.

Step 2. H normalizes N , so acts on $\square^n/N$ through H/N . The action is free on nonempty cells: if $h \cdot c = \nu \cdot c$ for some $\nu \in N$ then $\nu^{-1}h$ fixes the cell c , hence is a reflection or the identity, hence lies in N , so $h \in N$. By the covering theory of [5] (equivariant replacement of the free quotient; on the test side, el of the free quotient is the Borel construction), the quotient of a both-structures contractible object by a free action of the finite group H/N is a $B(H/N)$ in both structures — the argument of the free family of [3], applied verbatim with the contractible base $\square^n/N$ in place of $\square^n$. Finally $\square^n/H = (\square^n/N)/(H/N)$ since quotients are levelwise. $\square$

Remark 4.2. Theorem 4.1 subsumes the median-retraction family of [3]: for $H \leq \Sigma_n \times \langle z \rangle$ mixed, the reflections generate a subgroup fixing the diagonal-type locus, and the two-step argument applies with no parity condition on n . It also settles the five “edge” groups of dimension 4 found by the machine search — degenerate groups admitting *no* equivariant consensus pattern with odd negated classes — for which any one-step equivariant retraction provably fails: the two-step argument sidesteps the sign obstruction entirely, because all signs live in H/N and are absorbed by the covering step.

5. THE SEPARATION HALF

Theorem 5.1 (Separation at reflection-generated groups). *Let N be a mixed group with $N = \langle\langle \text{Refl}(N) \rangle\rangle$, on any reversal site, and suppose $\text{el}(\square^n/N)$ is not weakly contractible. Then an explicit test weak equivalence with target $\square^n/N$ that is not a type weak equivalence is attached to N .*

Proof (assembly). The witness machine is the E -resolution collage of [3] with $V = 1$, $G = N$, generalized to several N -orbits of strata: the body is $\square^n \times E_N$ with one horn per stratum, and the collage map $\bar{\Phi}_N$ projects to $\square^n/N$. The test side is the three-legs computation of [3], orbitwise: the sieve retracts horn-by-horn onto the strata ends; the cosieve covers are contractible with free N -action (nontrivial elements act freely on E_N); the comma splits into one piece per stratum orbit, each a classifying object of the stratum stabilizer on both sides. The type side is the nullity induction of [8, 3]: its stage-zero input is exactly certification, which holds by Theorem 3.1; its character hypothesis is Lemma 2.2; the slide and filler machinery is site-uniform [6, 5]. Non-contractibility of the target in the type structure follows from the assumption and the containment (U). $\square$

Hypothesis 5.2. *For every reflection-generated mixed $N \leq B_n$, the quotient $\square^n/N$ is not contractible in both structures simultaneously.*

Remark 5.3. If $\text{el}(\square^n/N) \not\simeq *$ the hypothesis holds by (U). If $\text{el}(\square^n/N) \simeq *$, then either $\square^n/N$ is not type-contractible — and is then itself an *object-level* witness, an even stronger form of separation [8] — or it is contractible in both structures, the only scenario the criterion must exclude. For $n \leq 3$

all relevant el's are computed and non-contractible (joins, homotopy colimits with $H_3 \neq 0$, the rank tables of [3]), so Hypothesis 5.2 holds and Theorem 1.2 is unconditional; the computations are site-invariant by the stratification invariance of [5].

Proof of Theorem 1.2. (1) is Theorem 4.1. (2): N is reflection-generated by construction, mixed by assumption; Theorem 5.1 (with Hypothesis 5.2) attaches a separation to N ; H/N contains no reflections (they all lie in N), hence acts freely on $\square^n/N$ and on the witness machine, and the covering-descent theorem of [5] transports the attached separation to H : the free quotient of the witness map is a test weak equivalence (Borel) and cannot be a type weak equivalence, since its equivariant cover would then be one. The 78/20 reproduction is the machine statement (Section 6). $\square$

6. MACHINE CROSS-CHECKS

The script `separation_criterion.py`, with the certificate engines of [5], verifies: the criterion against the family-by-family recursion of [3] on all 98 subgroups of B_3 — 78 agreement, 20 separation, zero mismatches; that $N = \langle\langle \text{Refl}(H) \rangle\rangle$ is generated by its own reflections in every case; that every reflection fixes a vertex; the certificate characterization of Theorem 3.1 on fourteen groups across $n = 2, 3, 4$ (all matching); the five degenerate dimension-4 groups with no one-step equivariant consensus pattern (Remark 4.2); and the certification of the five new dimension-4 primitive groups $V_{\text{even}}(4) \rtimes P$, $P \in \{C_4, K_4, D_4, A_4, S_4\}$. All scripts are archived in the artifact repository of the series

`https://github.com/r-shibusawa/cubical-strict-j`

(release v1.19.0, commit c39f71e, archived at DOI 10.5281/zenodo.22006175).

7. OUTLOOK

Three directions. First, Hypothesis 5.2: the collage decomposition shows that $\text{el}(\square^n/N) \simeq *$ would force the $\mathbb{F}_2$ -cohomology of N to decompose along its vertex stabilizers in a strong sense; we expect this to be impossible for mixed N and verifiable dimension by dimension. Second, the criterion is stated for reversal sites; on the connective sites without reversal there are no mixed subgroups to classify [6], so the criterion completes the isotropy theory of all standard cubical sites. Third, the slogan — separation is controlled by the reflection subgroup — suggests a synthesis with the theory of reflection arrangements: the fixed loci of $\text{Refl}(H)$ form an arrangement of subcubes, and the criterion says the two homotopy theories disagree exactly when the arrangement's symmetry cannot be disentangled from a fixed stratum. A precise dictionary with Coxeter-theoretic invariants is a natural sequel.

Data availability. The verification scripts are openly available in the repository above, archived on Zenodo: release v1.19.0, commit c39f71e, DOI 10.5281/zenodo.22006175.

Use of AI tools. An AI assistant was used in the preparation of this manuscript and its verification scripts; the author takes full responsibility for the content.

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